TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
System of inequalities: graph and shade each one; the overlap is the solution set.
A solution makes EVERY inequality true. Points on a dashed line are not solutions.
In a story: no negative amounts, so stay in the first quadrant.

1. Where is the solution set of $y < x + 1$ and $y < x + 4$?

Tip: both < → below both

A. below $y = x + 4$ only

B. below $y = x + 1$

C. between the lines

2. Is (3, 0) a solution of $y \geq -x + 2$ and $y \leq x - 1$?

Tip: check every inequality

F. Only the first is true

G. Only the second is true

H. Yes, both are true

3. A dashed line $y = \frac{1}{2}x - 1$ has the half-plane below it shaded. Which inequality is it?

> **Tip:** dashed + below → <

A. $y \leq \frac{1}{2}x - 1$

B. $y > \frac{1}{2}x - 1$

C. $y < \frac{1}{2}x - 1$

4. Together, $x \geq 0$ and $y \geq 0$ keep the solutions in which quadrant?

> **Tip:** both positive → Quadrant I

F. Quadrant I

G. Quadrant II

H. Quadrant III